P1 - S2 - 2018

1. Considere la siguiente igualdad:

$$1\cdot 5^1 + 2\cdot 5^2 + 3\cdot 5^3 + \cdots + n\cdot 5^n = \frac{5+(4n-1)\cdot 5^{n+1}}{16}$$

a) Demuestre, usando inducción matemática que dicha igualdad es válida para toda $n \geq 1$.

$n=1$ $\qquad 1\cdot 5^1 = \dfrac{5+(4\cdot 1-1)\cdot 5^{1+1}}{16}$

$5=5$ ✓

H.I.

$n=p \qquad 1\cdot 5^1 + 2\cdot 5^2 + 3\cdot 5^3 + \ldots + p\cdot 5^p = \dfrac{5+(4p-1)\cdot 5^{p+1}}{16}$

$n=p+1 \qquad 1\cdot 5^1 + 2\cdot 5^2 + 3\cdot 5^3 + \ldots + p\cdot 5^p + (p+1)\cdot 5^{(p+1)}$

Hay que Demostrar $= \dfrac{5+(4(p+1)-1)\cdot 5^{(p+1)+1}}{16}$

$\dfrac{5+(4p+4-1)\cdot 5^{p+2}}{16}$

$\dfrac{5+(4p+3)\cdot 5^{p+2}}{16}$

Demostracion

$1\cdot 5^1 + 2\cdot 5^2 + 3\cdot 5^3 + \ldots + p\cdot 5^p + (p+1)\cdot 5^{(p+1)}$

$\dfrac{5+(4p-1)\cdot 5^{p+1}}{16} + (p+1)\cdot 5^{(p+1)} \cdot \dfrac{16}{16}$, H.I.

$\dfrac{5+(4p-1)\cdot 5^{p+1} + 16(p+1)\cdot 5^{p+1}}{16}$

$$\frac{5+(4p-1)\cdot\boxed{5^{p+1}}+16(p+1)\cdot\boxed{5^{p+1}}}{16}$$

$$\frac{5+5^{p+1}(4p-1+16p+16)}{16}$$

$$\frac{5+5^{p+1}(20p+15)}{16}$$

$$\frac{5+(4p+3)\cdot 5^{p+2}}{16}$$

$$5^{p+1}\cdot 5^1$$

$$\frac{5+5^{p+1}\cdot 5^1(4p+3)}{16}$$

$$\frac{5+5^{p+2}(4p+3)}{16}$$

$$\sum_{i=1}^{n} n\cdot 5^n=\frac{5+(4n-1)\cdot 5^{n+1}}{16}$$

$$n,\ \frac{n!}{n},\ (n+1)^n$$

$$\sum_{i=1}^{p+1} a_n$$

$$\sum_{i=1}^{p} a_n + a_{n+1}$$

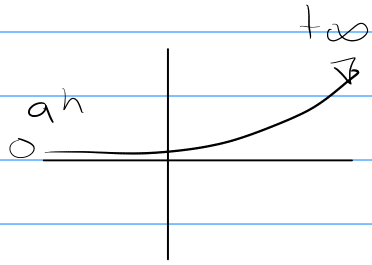
$$\sum_{i=1}^{p+1}\frac{5^n}{n}=\sum_{i=1}^{p}\frac{5^n}{n}+\frac{5^{n+1}}{n+1}$$

b) Si $\{S_n\} = \left\{\dfrac{5+(4n-1)\cdot 5^{n+1}}{16}\right\}$ corresponde a la sucesión de sumas parciales asociada a la serie $\sum_{k=1}^{\infty} k\cdot 5^k$, determine si dicha ~~serie~~ Sucesion converge o diverge. **(2 puntos)**

$$\sum_{k=1}^{\infty} a_n = \text{algo}$$

Trabajar con la igualdad

$$\sum_{k=1}^{\infty} k\cdot 5^k = \frac{5+(4n-1)\cdot 5^{n+1}}{16}$$



$$\lim_{n\to+\infty} \frac{5+(4\cdot+\infty-1)\cdot 5^{+\infty+1}}{16}$$

$$\frac{5+\infty\cdot\infty}{16} = +\infty \qquad \infty\infty \checkmark$$

$$\boxed{\therefore \text{ La serie diverge}}$$

4. Determine si cada una de las siguientes series convergen o divergen. Debe indicar los criterios aplicados en cada caso.

$a)\ \sum_{k=5}^{\infty} \frac{\arctan k + 4}{(k-4)^2}$ $\quad n = k$ **(4 puntos)**

Trigonometrico

Usar criterio de comparacion directa

Sen, Cos $-1 \leq x \leq 1$

$sen^2, \cos^2 \quad 0 \leq x \leq 1$

arctan $-\frac{\pi}{2} < x < \frac{\pi}{2}$

$$-\frac{\pi}{2} < \arctan(n) < \frac{\pi}{2}$$

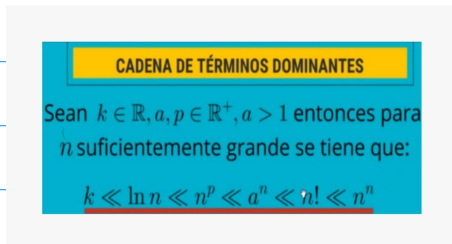
$$-\frac{\pi}{2}+4 < \arctan(n)+4 < \frac{\pi}{2}+4$$

$$\underbrace{\frac{-\frac{\pi}{2}+4}{(n-4)^2}} < \frac{\arctan(n)+4}{(n-4)^2} < \underbrace{\frac{\frac{\pi}{2}+4}{(n-4)^2}}$$

Elegir cualquiera

$n + \cancel{k}^{0} = n$, k constante

$\cancel{k}^{1} \cdot n = n$



$$\sum_{n=5}^{\infty} \frac{\frac{\pi}{2}+4}{(n-4)^2}$$

$$\frac{\pi}{2}+4 \sum_{n=5}^{\infty} \frac{1}{(n-4)^2}$$

$\frac{f(x)}{g(x)}$, $f(x) > g(x)$ $= +\infty$

$\sum_{n=5}^{\infty} \frac{1}{n^2}$, P serie $p = 2 > 1$ $\quad f(x) < g(x) = 0$

∴ Converge

$\frac{n^n}{n!} = +\infty$

$$\boxed{\therefore \frac{\arctan(n)+4}{(n-4)^2} \text{ Converge}}$$

$\frac{n!}{n^n} = 0$

Raiz

$$\lim_{n \to +\infty} \sqrt[n]{K} = 1$$

$$n! = n(n-1)(n-2)\ldots(n-K)!$$

$$\sqrt[n]{n^K} = 1$$

$(r)^n$ → Numero

$$\sqrt[n]{P(x)} = 1$$

$(a_n)^n$ → Expresion

$$\sqrt[n]{\frac{P(x)}{Q(x)}} = 1$$

$$\cancel{\sqrt[n]{n^n}} = n$$

b) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n \cdot \left(2+\frac{1}{n}\right)^{2n}}{e^n}$

$$\lim_{n \to +\infty} \sqrt[n]{|a_n|} < 1 \quad \text{Converge}$$

$$> 1 \quad \text{Diverge}$$

$$\lim_{n \to +\infty} \frac{\cancel{\sqrt[n]{|-1|^n}}^{\,1} \cdot \cancel{\sqrt[n]{}}\left[\left(2+\frac{1}{n}\right)^2\right]^{\cancel{n}}}{\cancel{\sqrt[n]{e^n}}}$$

$$\sqrt[n]{\frac{a \cdot b}{c}} = \frac{\sqrt[n]{a} \cdot \sqrt[n]{b}}{\sqrt[n]{c}}$$

$$\lim_{n \to +\infty} \frac{\left(2+\frac{1}{n}\right)^2}{e}$$

$$a^{xy} = (a^x)^y = (a^y)^x$$

$$\frac{\left(2+\cancel{\frac{1}{\infty}}^{\,0}\right)^2}{e}$$

$$2^{3x} = (2^3)^x = (2^x)^3$$

$$\frac{2^2}{e} = \frac{4}{e} > 1$$

$$\boxed{\therefore \text{ Diverge}}$$

$a_n \qquad b_n$

c) $\displaystyle\sum_{n=1}^{\infty}\left[\frac{(-1)^{3n}\cdot n^2}{n^3+1} - \frac{1}{3^{-n+4}}\right]$

$$\sum_{n=1}^{\infty} \frac{(-1)^{3n}\cdot n^2}{n^3+1} - \sum_{n=1}^{\infty} \frac{1}{3^{-n+4}}$$

$$\sum_{n=1}^{\infty} \frac{1}{3^{-n}\cdot 3^4}$$

$$\frac{1}{3^4}\sum_{n=1}^{\infty} \frac{1}{3^{-n}}$$

$2^2 = 2^2$

$2^{-2} = \frac{1}{2^2}$

$$\frac{1}{3^4}\sum_{n=1}^{\infty} \frac{\frac{1}{1}}{\frac{1}{3^n}}$$

$$\frac{1}{3^4}\sum_{n=1}^{\infty} \left(3\right)^{n}$$

r^n

$|r| < 1$
Converse
$|r| > 1$
Diverge

$|r| = |3| = 3 > 1$

$\therefore$ Diverge

$$\boxed{\begin{array}{l}\therefore \\ \text{c) } \displaystyle\sum_{n=1}^{\infty}\left[\frac{(-1)^{3n}\cdot n^2}{n^3+1} - \frac{1}{3^{-n+4}}\right] \\ \text{Diverge}\end{array}}$$

5. Si la función $f : [2, +\infty[\longrightarrow \mathbb{R}$ tal que $f(x) = \dfrac{1}{x(\ln x)^2}$ es continua, positiva y decreciente, determine si la serie $\displaystyle\sum_{k=2}^{\infty} (-1)^{k+1} \cdot \frac{1}{k(\ln k)^2}$ diverge, converge absolutamente o converge condicionalmente. **(4 puntos)**

$$\sum_{n=1}^{\infty} (-1)^n \cdot \frac{1}{n^2}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2}, \quad p \text{ serie}, \quad p = 2 > 1$$

$\therefore$ Converge

$$\sum_{n=1}^{\infty} (-1)^n \cdot \frac{1}{n^2} \qquad \boxed{\text{Converge Absolutamente}}$$

$$\sum_{n=1}^{\infty} (-1)^n \cdot \frac{1}{n^1}$$

$$\sum_{n=1}^{\infty} \frac{1}{n}, \quad p \text{ serie}, \quad p = 1 \leq 1$$

Diverge

$\to f(x) \searrow \qquad -\dfrac{1}{x^2}$, Decreciente

$\lim\limits_{x \to +\infty} a_n = 0 \qquad \lim\limits_{x \to +\infty} \dfrac{1}{x} = 0$

$\therefore$ Converge Condicionalmente

5. Si la función $f : [2, +\infty[\longrightarrow \mathbb{R}$ tal que $f(x) = \dfrac{1}{x(\ln x)^2}$ es continua, positiva y decreciente, determine si la serie $\displaystyle\sum_{k=2}^{\infty} (-1)^{k+1} \cdot \frac{1}{k(\ln k)^2}$ diverge, converge absolutamente o converge condicionalmente. **(4 puntos)**

$n = k$

$$\sum_{n=2}^{\infty} \frac{1}{x(\ln(x))^2}$$

$$\int_2^{+\infty} \frac{1}{x(\ln(x))^2}$$

$$\int \frac{1}{x(\ln(x))^2} dx \qquad u = \ln(x) \quad du = \frac{1}{x} dx$$

$$\int \frac{1}{u^2} du \rightarrow \int u^{-2} = \frac{u^{-2+1}}{-2+1}$$

$$\frac{u^{-1}}{-1} = \frac{-1}{u}$$

$$\frac{-1}{\ln(x)} \Big|_2^{+\infty}$$

$$\lim_{n \to +\infty} \frac{-1}{\cancel{\ln(+\infty)}} \; 0 - -\frac{1}{\ln(2)}$$

$$\frac{1}{\ln(2)}$$

$$\boxed{\therefore \text{ Converge Absolutamente}}$$

6. Asuma que la serie $S = \sum_{k=1}^{\infty} \frac{(-1)^k \cdot 4^k}{(2k+1)\cdot(2k+1)!}$ es convergente. Determine el menor valor para n de manera que S_n aproxime a S con un error menor a 10^{-6}. **(3 puntos)**

$N = 5$

(10^{-6})

N Respuesta	N+1 Probar	a_n Serie	$< 0{,}000001$ $< 0{,}000001$
4	5	$\frac{4^5}{(2\cdot 5+1)(2\cdot 5+1)!}$	$\approx 0{,}000002$ $\not< 0{,}000001$
5	6	$\frac{4^6}{(2\cdot 6+1)(2\cdot 6+1)!}$	$\approx 0{,}00000005$ $< 0{,}000001$

$\boxed{N=5}$